

Smoke, Alcohol and LPG-Detection Alarm

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Presented here is a circuit for raising an alarm on detecting smoke or LPG cooking gas leakage, or even alcohol vapours in breath. This is achieved by using a basic unit with different sensors for

smoke, LPG and alcohol. So, different alarms can be made by simply changing the input sensor.

Circuit and working

Fig. 1 shows circuit diagram of the smoke alarm. The basic unit employs dual op-amp LM358 (IC1), transistor



T1 and timer NE555 (IC2), configured in astable mode. Selection of MQ* sensor is based on the purpose for

which the alarm circuit is being designed. For the smoke alarm circuit, you will need MQ2 sensor. Similarly, for detection of alcohol, MQ3 sensor is used, while MQ6 sensor is used for detecting LPG cooking gas.

MQ* sensor has six pins. Heater filament, between pins H-H, is directly connected to 5V and ground. Two sets of input/output (I/O) pins A-A and B-B are present. Select any

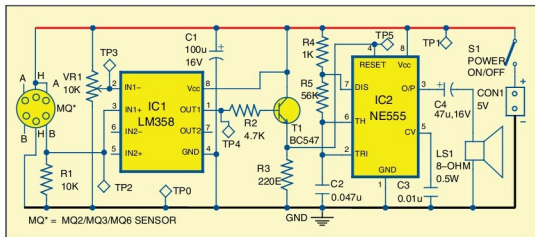


Fig. 1: Circuit diagram of the alarm circuit

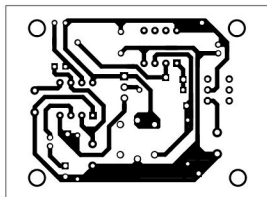


Fig. 2: Actual-size PCB pattern of the alarm circuit

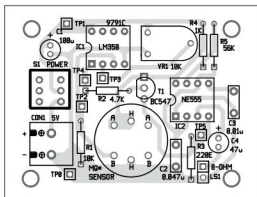


Fig. 3: Component layout of the PCB

Test Points

Test point	Details
TP0	0V (GND)
TP1	5V when switch S1 is on
TP2	Around 1.8V when no alcohol breath
TP3	2.6V set using VR1
TP4	Low when no alcohol breath and high when there is alcohol breath
TP5	Low when no alcohol breath and high when there is alcohol breath

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